

Impact of Acquisition Function on Implicit Neural Representation Training

Szymon Kiełtyka

SZYMON.KIELTYKA.STUD@PW.EDU.PL

Sebastian Botero Leonik

SEBASTIAN.BOTERO_LEONIK.STUD@PW.EDU.PL

Warsaw University of Technology, Poland

Abstract

Standard active learning loops for implicit neural representations typically rely on predictive variance, acquiring samples independently while ignoring parameter constraints and spatial redundancy. While MaxVar sampling is computationally cheap, it frequently over-samples clustered regions, leading to suboptimal sample efficiency during training. We systematically compare advanced active acquisition strategies (incorporating information-theoretic grounding via FisherEIG and determinantal batch diversity via BatchDPP) against standard continuous baselines. On the Gaussian Random Field benchmark using an acquisition batch size of 1024, the FisherEIG strategy achieves a final reconstruction peak signal-to-noise ratio (PSNR) of 40.68, outperforming the active MaxVar baseline (39.53 PSNR) and the passive Sobol sequence (37.87 PSNR). Furthermore, we show that the advantage of these methods relative to MaxVar depends heavily on the acquisition batch size (k). BatchDPP significantly outperforms MaxVar at large batch sizes ($k = 128, 1024$) by mitigating the spatial redundancy inherent in MaxVar’s greedy top- k selection. Conversely, in the low-batch regime ($k = 2, 16$), the marginal reconstruction gains of BatchDPP and FisherEIG do not justify their submodular optimization overhead, leaving MaxVar as the most efficient choice.

1. Introduction

Active learning (Pan et al., 2022) optimizes sample efficiency by strategically querying an oracle, which is critical when data acquisition is expensive. While MaxVar sampling serves as a highly competitive baseline, it selects points independently, often wasting computational budget on spatially redundant queries. Recent approaches attempt to overcome these limitations through three distinct directions: information-theoretic grounding (FisherEIG, Jiang et al. (2025)) to actively constrain model parameters, determinantal point processes (BatchDPP, Biyik et al. (2019)) to enforce spatial batch diversity, and gradient-based methods (GradAcq, Settles et al. (2007)) that target regions where new data will most strongly update model weights. However, it remains a question whether the empirical gains of these sophisticated active strategies justify their significant computational overhead when compared to simple variance sampling or strong passive baselines like Sobol. In this work, we systematically evaluate four active acquisition strategies (MaxVar, FisherEIG, BatchDPP, GradAcq) against four baselines across both analytical benchmarks and a physically motivated oracle. Using a SIREN (Sitzmann et al., 2020) architecture, we characterize the specific conditions under which they meaningfully outperform standard variance sampling in terms of reconstruction quality and sample efficiency.

2. Contributions

- We provide a comprehensive empirical benchmark tracking reconstruction accuracy (via NRMSE and PSNR) for eight unique active learning acquisition strategies evaluated across four distinct target datasets (Ackley, Branin, Gaussian Random Field, and Rastrigin) under fully standardized experimental controls.
- We evaluate the direct performance impact of varying the batch update budget k across four logarithmic scales ($k \in \{2, 16, 128, 1024\}$), demonstrating that increasing parallel step budgets tends to yield monotonic improvements in target reconstruction fidelity rather than suffering from information redundancy.
- We establish that computationally intensive active selection strategies (MaxVar, FisherEIG, BatchDPP, and GradAcq) do not consistently outperform passive baselines across all datasets, showing for instance that standard uniform random sampling achieves matching or superior peak signal-to-noise ratios on the Branin dataset.

3. Related Works

3.1 Implicit Neural Representations.

The representation of signals as continuous, differentiable functions parameterized by neural networks has emerged as a robust paradigm for modeling complex data. (Sitzmann et al., 2020) While standard architectures often rely on ReLU-based multilayer perceptrons (MLPs), these models frequently struggle to capture fine details and higher-order derivatives because their second derivatives are zero everywhere. SIREN (Sinusoidal Representation Networks) addresses this limitation by using periodic activation functions. (Sitzmann et al., 2020) This approach allows for the high-fidelity representation of natural signals such as images, audio, and video, while remaining well-behaved under supervision from derivatives, which is essential for solving complex boundary value problems.

3.2 Active Learning Strategies.

Active Learning (AL) aims to achieve maximal model performance with a constrained data budget by iteratively selecting only the most informative samples to be labeled. General active learning approaches are typically categorized into two classes: representativeness-based methods, which focus on increasing the diversity of the training set (e.g., core-sets), and informativeness-based methods. (Pan et al., 2022) Informativeness-based strategies measure the uncertainty of each potential data point and select candidates that are expected to yield the highest information gain or the greatest reduction in model variance. In the context of signal reconstruction, these principled selection strategies are designed to outperform passive baselines like random sampling by identifying regions where the current model lacks sufficient detail.

4. Method

We model the target signals using a SIREN neural network architecture with 4 hidden layers of 256 neurons, obtaining predictive uncertainty estimates via Monte Carlo Dropout (Gal

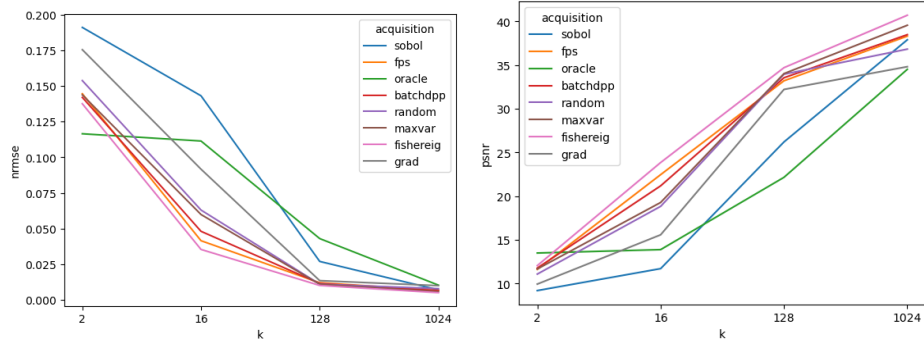


Figure 1: Evaluation of reconstruction accuracy via Normalized Root Mean Squared Error (NRMSE, left) and Peak Signal-to-Noise Ratio (PSNR, right) as a function of batch size k on the Gaussian Random Field (GRF) target landscape. The horizontal axes display the batch update budgets ($k \in \{2, 16, 128, 1024\}$), while the vertical axes chart the respective performance metrics, with distinct colored lines mapping individual active and passive acquisition strategies. Both metrics reveal a clear and consistent improvement as the batch budget increases, characterized by a monotonic decrease in NRMSE and a corresponding increase in PSNR demonstrating that larger batch sizes facilitate significantly better global convergence and surface reconstruction fidelity on this dataset

and Ghahramani, 2016). We perform 10 learning rounds. In each active learning round we perform a single training epoch on the set of previously sampled points then the acquisition function selects a batch of k coordinates to add to the training set. The training is performed using the Adam optimizer with PyTorch’s default configuration. We assess four principled active strategies :

- **MaxVar** - Top- k coordinates by $\sigma^2(x)$. Per-point evaluation, no batch coordination.
- **FisherEIG** - Maximise log-det of Fisher Information Matrix (D-optimal). Greedy submodular selection with $(1 - 1/e)$ guarantee. Requires LaplaceLL for analytical FIM.
- **BatchDPP** - Determinantal Point Process: kernel $L_{ij} = \sigma(x_i)\sigma(x_j) \cdot \text{sim}(\phi(x_i), \phi(x_j))$. Quality-diversity tradeoff via α parameter.
- **GradAcq** - Expected gradient length: $4 \cdot \sigma^2(x) \cdot \|\nabla_{\theta} f(x)\|^2$. Targets uncertain coordinates where new data would most move the parameters.

These are contrasted against four baselines: **Random** sampling, **Sobol** sequences, **Farthest Point Sampling** and an impractical theoretical ceiling, **Oracle Error**.

5. Experiments

We evaluate 8 acquisition functions across four datasets: Ackley, Branin, Gaussian Random Field and Rastrigin. We conduct a grid search over acquisition step batch sizes $k \in$

dataset acquisition	Ackley	Branin	GRF	Rastrigin
BatchDPP	18.86	15.44	38.46	25.47
FisherEIG	19.05	15.45	40.68	25.26
GradAcq	19.36	15.34	34.79	25.11
MaxVar	19.04	15.45	39.53	24.73
FPS	19.02	15.45	38.29	23.15
Oracle	16.44	12.83	34.49	26.90
Random	19.16	15.46	36.80	23.65
Sobol	2.01	13.95	37.87	7.59

Table 1: Peak Signal-to-Noise Ratio (PSNR) performance of eight acquisition functions across four benchmark datasets. Rows represent individual acquisition methods and columns correspond to the benchmark functions, with the highest value per column highlighted in bold. The grad and fishereig strategies achieve the highest overall performance on Ackley and GRF respectively, while the baseline sobol sequence underperforms substantially.

$\{2, 16, 128, 1024\}$. All experiments are run with a fixed initialization budgets $k_0 = 16$ and random seed of 2137. To assess reconstruction quality, we track two metrics at each optimization step: **PSNR** and **NRMSE**.

6. Conclusions

The results demonstrate that while BatchDPP and FisherEIG consistently improve sample efficiency over the Sobol baseline and outperform the Random baseline on specific landscapes like GRF and Rastrigin, their advantage relative to MaxVar depends heavily on both the acquisition batch size (k) and the underlying topography of the target function. Table 1 summarizes the final reconstruction metrics across all evaluated environments, highlighting this complex performance crossover.

Evaluating these metrics reveals highly dataset-dependent behaviors. On the Rastrigin dataset, BatchDPP clearly outperforms MaxVar by enforcing spatial diversity, bypassing the redundancy of MaxVar’s greedy top- k selection. Conversely, in environments like Gaussian Random Fields or Branin, MaxVar proves unexpectedly resilient, matching or exceeding BatchDPP without explicit diversity constraints. Notably, it is FisherEIG that achieves the highest overall performance on the GRF dataset.

Furthermore, performance trajectories on the GRF dataset (Figure 1) show that in the low-batch regime ($k = 2$ and 16), reconstruction gains of BatchDPP and FisherEIG over MaxVar are strictly marginal. Because spatial redundancy is naturally constrained at smaller step sizes, these slight predictive improvements fail to justify the heavy computational overhead of submodular optimization. Therefore, for smaller values of k on this landscape, MaxVar emerges as the most pragmatic, cost-effective choice.

References

- Erdem Biyik, Kenneth Wang, Nima Anari, and Dorsa Sadigh. Batch active learning using determinantal point processes. *ArXiv*, abs/1906.07975, 2019. URL <https://api.semanticscholar.org/CorpusID:195069132>.
- Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: representing model uncertainty in deep learning. In *Proceedings of the 33rd International Conference on International Conference on Machine Learning - Volume 48*, ICML’16, page 1050–1059. JMLR.org, 2016.
- Wen Jiang, Boshu Lei, and Kostas Daniilidis. Fisherrf: Active view selection and mapping with radiance fields using fisher information. In Aleš Leonardis, Elisa Ricci, Stefan Roth, Olga Russakovsky, Torsten Sattler, and Gül Varol, editors, *Computer Vision – ECCV 2024*, pages 422–440, Cham, 2025. Springer Nature Switzerland. ISBN 978-3-031-72624-8.
- Xuran Pan, Zihang Lai, Shiji Song, and Gao Huang. Activenerf: Learning where to see with uncertainty estimation. In Shai Avidan, Gabriel Brostow, Moustapha Cissé, Giovanni Maria Farinella, and Tal Hassner, editors, *Computer Vision – ECCV 2022*, pages 230–246, Cham, 2022. Springer Nature Switzerland. ISBN 978-3-031-19827-4.
- Burr Settles, Mark Craven, and Soumya Ray. Multiple-instance active learning. In J. Platt, D. Koller, Y. Singer, and S. Roweis, editors, *Advances in Neural Information Processing Systems*, volume 20. Curran Associates, Inc., 2007. URL https://proceedings.neurips.cc/paper_files/paper/2007/file/a1519de5b5d44b31a01de013b9b51a80-Paper.pdf.
- Vincent Sitzmann, Julien Martel, Alexander Bergman, David Lindell, and Gordon Wetzstein. Implicit neural representations with periodic activation functions. In H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 7462–7473. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/file/53c04118df112c13a8c34b38343b9c10-Paper.pdf.

Appendix A. Additional table and plots

dataset acquisition	Ackley	Branin	GRF	Rastrigin
BatchDPP	0.1165	0.1692	0.0066	0.0533
FisherEIG	0.1141	0.1691	0.0051	0.0546
GradAcq	0.1100	0.1713	0.0100	0.0556
MaxVar	0.1141	0.1690	0.0058	0.0580
FPS	0.1144	0.1690	0.0067	0.0696
Oracle	0.1540	0.2286	0.0104	0.0452
Random	0.1126	0.1690	0.0080	0.0657
Sobol	0.8107	0.2010	0.0070	0.4174

Table 2: The same kind of table as in the paper but with data for NRMSE. More decimal points are displayed to properly capture the differences

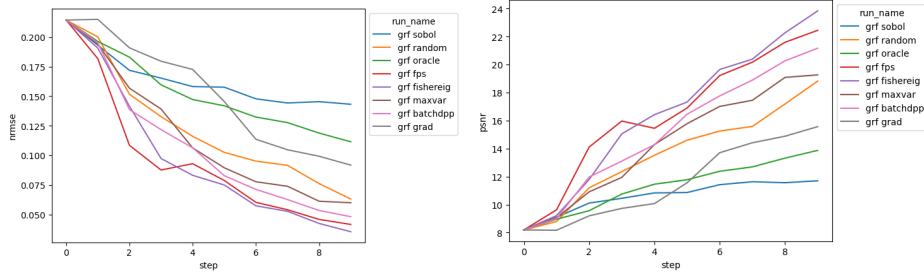


Figure 2: Plots showing metric evaluation for subsequent steps of the training process. $k=16$ and the oracle is Gaussian Random Fields. Plots display the metrics NRMSE(left) and PSNR(right)